

RSM 8224
Analytical Insights Using Accounting and Financial Data
Group Project:
Assess the Relation between ESG Performance and Financial Returns

Slides Due Date: April 17, 2026 (9:59 am before class, via email)

Report Due Date: April 20, 2026 (11:59 pm, via Quercus)

Please submit **one research report** (PDF) for each group

Introduction

This group project investigates whether firms that perform well on Environmental, Social, and Governance (ESG) dimensions also perform well financially. Is ESG performance financially material, and does it matter for investors? Does doing good mean doing well?

The answer depends on how you define “matter.” Recent regulatory developments – such as the EU Corporate Sustainability Reporting Directive (CSRD) – have introduced the concept of *double materiality*, which distinguishes two channels through which ESG issues relate to a firm:

- **Financial materiality:** ESG issues affect a firm’s financial performance and value through physical risks and transition risks. For example, an environmental regulation that raises production costs for a chemical manufacturer is financially material. This perspective asks: does ESG performance predict earnings or returns?
- **Impact materiality:** A firm’s operations affect the environment and society, irrespective of whether those effects feed back into the firm’s own financial performance. A factory’s carbon emissions harm the environment regardless of whether carbon pricing exists. This perspective asks: does the firm’s ESG performance reflect real-world positive or negative externalities?

In this group project, you will use ESG ratings from the LSEG (Refinitiv) ESG database on WRDS, merged with accounting data from Compustat and stock return data from CRSP, to address these questions empirically.

Note: This is a group project. You have flexibility to design and organize your research report for the questions. You may discuss your ideas with AI and use AI to help with your coding, but the final report cannot be generated by AI.

Data Sources

All data will be obtained from WRDS. You will use three primary datasets: LSEG (Refinitiv) ESG scores, Compustat financial statement data, and CRSP stock return data.

ESG ratings attempt to quantify how well a firm manages environmental, social, and governance risks and opportunities. Refinitiv (LSEG) ESG scores are available on WRDS (`tr_esg`) and cover over 10,000 companies globally. The scores are structured hierarchically: overall ESG

score is a composite rating (0–100) aggregating all three pillars (E/S/G). Please use the overall score in your primary analysis.

Merge the ESG data with CRSP and Compustat. Use the CRSP–Compustat Merged (CCM) link table to map `gvkey` to `permno`, following our example codes in the individual assignment.

Next, merge LSEG (Refinitiv) ESG data to Compustat via CUSIP. A U.S. ISIN has the format "US" + 9-digit CUSIP + 1 check digit. Compustat's `cusip` field stores the first 8 digits of the CUSIP. Extract characters 3–10 from the ISIN to obtain the 8-digit CUSIP, then match to Compustat. Reference on WRDS data merging:

<https://wrds-www.wharton.upenn.edu/pages/wrds-research/database-linking-matrix/>

Sample

Your sample should cover fiscal years 2013–2023. Apply the following filters:

- Domestic common stocks: `shrcd IN (10, 11), exchcd IN (1, 2, 3)` (NYSE, AMEX, NASDAQ).
- Standard Compustat format: `indfmt='INDL', datafmt='STD', consol='C', popsrc='D'`.
- Valid CCM link: `linktype IN ('LC', 'LU', 'LS'), linkprim IN ('C', 'P')`, and `datadate` within the valid link window.
- Non-missing ESG score, positive total assets (`at > 0`), and at least 12 months of CRSP return data after fiscal year end.

Your final sample should include annual firm characteristics, stock returns, and ESG data in a panel format.

Part I:

Describe your final sample. Report basic descriptive statistics for your variables.

Present the distribution of ESG scores across industries and over time. Are ESG scores improving on average? Which industries are doing good?

Part II:

What determines firms' ESG performance? Pick firm characteristics of your interest and explore the relations with ESG performance.

Part III:

Does doing good mean doing well? Report the regression results of financial performance on lagged ESG scores. Does ESG predict future ROA after controlling for observable firm characteristics? Does ESG predict stock returns either unconditionally or after controlling for earnings news? Please discuss your findings in the framework of financial materiality and impact materiality.

Part IV:

Conduct industry-level analysis. Which industries are “double materiality winners” (high ESG and high financial performance)? Which industries exhibit a tension (high ESG but low returns, or low ESG but high returns)? What does this imply for how ESG-conscious investors should allocate investment?

Lastly, based on your analysis, recommend one industry where you believe an investor can achieve both competitive financial returns and positive ESG impact. Provide descriptive statistics on the firms in that industry and explain your selection.

Submission

For the class presentation, each group will have 10 – 12 minutes, so please prepare a maximum of 15 slides. Focus on your most interesting and surprising findings rather than walking through every table. Visual clarity and storytelling matter.

Submit a research report in PDF format (**max 15 pages**, excluding appendices, font size 12). The report should read as a professional research document, and you may structure as follows:

- Introduction: Motivate the research question and preview your main findings (≤ 2 pages).
- Data and Sample: Describe data sources, sample construction, variable definitions, and summary statistics (≤ 2 pages).
- Empirical Analysis: Present your results for each part with results (figures and tables) and interpretation.
 - Part I
 - Part II
 - Part III
 - Part IV
- Conclusion: Summarize key takeaways, discuss limitations (1 page).
- References: A list of cited works.
- Appendix: Any additional results or information. Codes are NOT needed.

Academic resources that may be useful:

- Berg, F., Koelbel, J.F., and Rigobon, R. (2022). Aggregate Confusion: The Divergence of ESG Ratings. *Review of Finance*, 26(6), 1315–1344.
- Bolton, P. and Kacperczyk, M. (2021). Do Investors Care about Carbon Risk? *Journal of Financial Economics*, 142(2), 517–549.
- Christensen, H.B., Hail, L., and Leuz, C. (2021). Mandatory CSR and Sustainability Reporting: Economic Analysis and Literature Review. *Review of Accounting Studies*, 26, 1176–1248.
- Khan, M., Serafeim, G., and Yoon, A. (2016). Corporate Sustainability: First Evidence on Materiality. *The Accounting Review*, 91(6), 1697–1724.
- Pastor, L., Stambaugh, R.F., and Taylor, L.A. (2022). Dissecting Green Returns. *Journal of Financial Economics*, 146(2), 403–424.
- Serafeim, G., & Yoon, A. (2023). Stock price reactions to ESG news: The role of ESG ratings and disagreement. *Review of Accounting Studies*, 28(3), 1500-1530.
- Starks, L. T. (2023). Presidential address: Sustainable finance and ESG issues—Value versus values. *Journal of Finance*, 78(4), 1837-1872.